

Technology #08: NIR Photovoltaic Power Delivery for Implantable Devices

Technology Domain: Optics + Photonics + Power Electronics

Mechanism: External 940nm LED → scalp/skull tissue → implanted GaAs PV cell → power conditioning → sensor/load

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Tissue attenuation at 5cm depth prevents sufficient power delivery (>1mW/cm² required at PV surface)

Next Experiment: NIR source + PV cell bench test

Buyer Receives: 940nm optical power delivery concept + verified fluence (T2-CONFIRMED); Computational model (PyTissueOptics v2.0.1, DCC-Lab independent verification); Optical properties specification (Jacques 2013); Bench experiment protocol (LED + phantom + PV); External precedent: PMC5646820 confirms subcutaneous PV IR harvesting; This engineering dossier with optical-specific design chain

Buyer Must Build: Implantable PV cell packaging (biocompatible, optically transparent at 940nm); Thermal management design (LED heating + PV heating); External LED patch design (wearable, chronic use); Power conditioning circuit (implantable); Regulatory submission (PMA — optical implant); Clinical validation evidence

Transfer Posture: Sponsored Validation

Target Buyer: Implantable device company; neuroshunt manufacturer

CONFIDENTIAL. See full engineering dossier for details.